

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	30 September 2026
Team ID	SWTID-2026-1542
Project Name	AnswerBot – AI-powered FAQ and Customer Support Automation
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form (name, email, password) Unique email validation Password hashing using bcrypt before storage
FR-2	User Authentication	Login using email and password JWT token generation on successful login Access to protected routes using Bearer token
FR-3	FAQ Management (CRUD)	Create a new FAQ (question, answer, category) View / list all published FAQs Update own FAQ records Delete own FAQ records
FR-4	AI FAQ Generation	Generate question, answer and category from a topic using Google Gemini Review AI-generated FAQ before saving Save generated FAQ to the database
FR-5	FAQ Search	Keyword and semantic search using query parameters Filter FAQs by category
FR-6	Category Management	Create and organise FAQ categories View number of FAQs per category
FR-7	Role-Based Access Control	Admin: manage users, all FAQs and categories Content Creator: manage own FAQs and use AI generation Public User: read-only access to published FAQs and search

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Consistent, well-structured REST endpoints with clear JSON request/response formats and meaningful error messages, so the API is easy to test in Postman/ThunderClient and easy for any frontend to consume.
NFR-2	Security	Stateless JWT authentication and route guarding, bcrypt password hashing, role-based authorization so only the original author (or Admin) can modify an FAQ, environment variables (dotenv) for secrets, and sanitized error responses that do not leak database details.
NFR-3	Reliability	Centralized error handling with standard error responses, Mongoose schema validation before persistence, and graceful handling of database or AI-service failures without crashing the server.
NFR-4	Performance	Indexed MongoDB queries for fast FAQ search and retrieval; lightweight Express middleware; AI generation requests handled asynchronously so other requests are not blocked.
NFR-5	Availability	Stateless backend that can be restarted or redeployed without losing user sessions; database connection is retried on failure; can be hosted on a cloud platform with a managed MongoDB service (e.g. MongoDB Atlas) for high uptime.
NFR-6	Scalability	Modular MVC structure (routes, controllers, models, services) and stateless design allow horizontal scaling behind a load balancer; MongoDB supports replication and sharding as data grows.